Common Random Number applied to examples

Sabrina Sixta

June 24, 2025

1 Convergence rate of CRN and backward processes are the same

Define a Markov chain $\{X_n\}_{n\geqslant 1}$ in $(\mathcal{X}, \mathcal{F})$ such that $X_n = f_{\theta_n}(X_{n-1})$, where $\{\theta_n\}_{n\geqslant 1}$ are i.i.d. random variables on some measurable space Θ and random measurable mappings $f_{\theta_n} : \mathcal{X} \mapsto \mathbb{R}$. The set of random functions $f_{\theta_1}, f_{\theta_2}, \ldots$ is called an iterated function system. Any time-homogeneous Markov chain can be represented as an iterated function system [10].

The iterated function system defines the forward and backward processes. The forward process, $\{X_n\}_{n\geqslant 1}$, which is a Markov chain, is defined as follows,

$$X_n = f_{\theta_n}(f_{\theta_{n-1}}(\ldots f_{\theta_1}(X_0)))$$

The backward process, $\{\tilde{X}_n\}_{n\geqslant 1}$, which is not a Markov chain [9] with respect to filtration for $X_0, \ldots, X_n$ but tends to converge pointwise to a limit, is defined as follows,

$$\tilde{X}_n = f_{\theta_1}(f_{\theta_2}(\ldots f_{\theta_n}(X_0)))$$

Figure 1 graphs the forward and backward process of an autoregressive normal process using the same random mappings. The point towards which the backwards process converges is itself

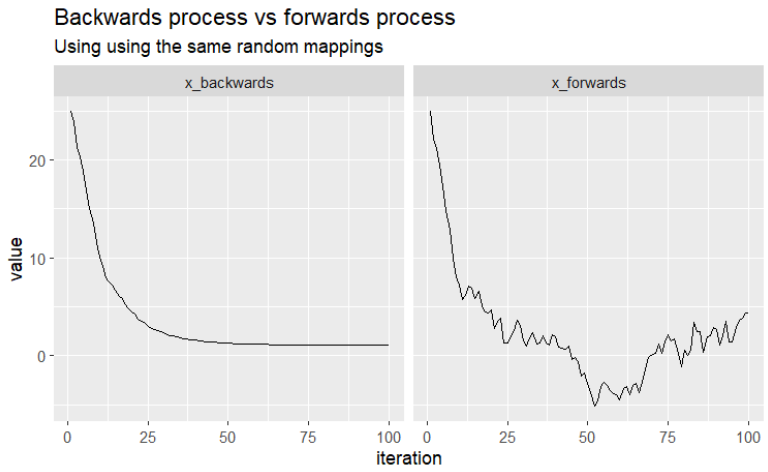


Figure 1: The backwards, $\{\tilde{X}_n\}_{n\geqslant 1}$, and forwards, $\{X_n\}_{n\geqslant 1}$, process of the autoregressive normal system where $X_n, \tilde{X}_n$ are simulated using the same random mappings, $X_n = 0.9X_{n-1} + Z_n, Z_n \sim N(0.1)$ and $X_0 = 25$.

random.

Despite the difference in behaviour of the two processes, the marginal distributions of the forward and backward processes are the same, $X_n \stackrel{d}{=} \tilde{X}_n$, so the backwards process is sometimes used when studying convergence properties of the forwards Markov chain [4, 10, 9].

Theorem 1.1. *The expected distance between the nth iteration of the two forward processes where $X_0 = x$ and $X_0' = f_{\theta_{n+1}}(x)$ is equal to the expected distance between the nth and $(n+1)$th iteration of the backwards process, $\tilde{X}_n$.*

$$E[d(X_n, X_n') \mid X_0 = x, X_0' = f_{\theta_{n+1}}(x)] = E[d(\tilde{X}_n, \tilde{X}_{n+1}) \mid \tilde{X}_0 = x]$$

Proof.

$$
\begin{aligned}
&E[d(X_n, X_n') \mid X_0 = x, X_0' = f_{\theta_{n+1}}(x)] \\
&= E[d[f_{\theta_n}(f_{\theta_{n-1}}(\dots f_{\theta_1}(x)\dots)), f_{\theta_n}(f_{\theta_{n-1}}(\dots f_{\theta_1}(f_{\theta_{n+1}}(x))\dots))] \mid X_0 = x] \\
&= E[d[f_{\theta_1}(f_{\theta_2}(\dots f_{\theta_n}(x)\dots)), f_{\theta_1}(f_{\theta_2}(\dots f_{\theta_n}(f_{\theta_{n+1}}(x))\dots))] \mid \tilde{X}_0 = x] \\
&= E[d(\tilde{X}_n, \tilde{X}_{n+1}) \mid \tilde{X}_0 = x]
\end{aligned}
$$

□

2 Sufficient conditions for when CRN converges a.s. when there is convergence in distribution

The following is a theorem that we will use to prove that random variables simulated according to common random number converge almost surely. We've adjusted the theorem notation to describe distribution functions.

Proposition 2.1 (Theorem 2 of [1])**.** *Let $\{F_n\}_{n\geqslant 0}$ be a sequence of mappings. If*

1. *$\{F_n\}_{n\geqslant 0}$ is a sequence of injective mappings on a metric space $(\mathbb{R}^d, ||^2)$ and taking values in a locally compact metric space $([0,1], ||)$,*
2. *$F_n \to F$ uniformly on $\mathbb{R}^d$,*
3. *F^{-1} is a continuous mapping on $[0,1]$*

Then $F_n^{-1} \to F^{-1}$ uniformly on every compact set K_0 contained in $N_0 = Interior[0,1] \cap_{n\geqslant 1} Im(F_n)$

Theorem 2.2 is a general theorem, but we are only interested in the theorem applied to distribution functions. Many of the requirements are already fulfilled for distribution functions (for example if $F_n \to F$ and F is continuous then the convergence is uniform (see example

14.8 of [2], theorem 1.1 of this), the metric space $([0,1],\lambda)$ is already locally compact, if F^{-1} is continuous then F is continuous Why does Google always send me to mathStackExchange??? etc.), so we can rewrite theorem 2.2 as follows.

Proposition 2.2 (Theorem 2 of [1] adjusted for distribution functions)**.** *Let $\{F_n\}_{n\geqslant 0}$ be a sequence of mappings. If*

1. *$\{F_n\}_{n\geqslant 0}$ is a sequence of strictly increasing distribution functions on a metric space $(\mathbb{R}^d,||^2)$ and taking values in $([0,1],||)$,*
2. *$F_n \to F$ on $\mathbb{R}^d$*
3. *F is a continuous, invertible mapping on $[0,1]$*

Then $F_n^{-1} \to F^{-1}$ uniformly on every compact set $K_0 \in (0,1)$.

Definition 2.1 (Uniform convergence of functions)**.** The given the sequence of functions $F_n : (\mathbb{R}^d,||^2) \to ([0,1],||)$ and the target function $F : (\mathbb{R}^d,||^2) \to ([0,1],||)$, $F_n \to F$ uniformly if for all $\epsilon > 0$, there exists an $N \in \mathbb{N}$ such that for any $n \geqslant N$ and $x \in \mathbb{R}^d$,

$$|F_n(x) - F(x)| \leqslant \epsilon$$

This theorem implies the following result. Namely let F be a target distribution function and

1. $\{X_n(u)\}_{n\geqslant 1}$ be a sequence of random variables with $X_n(u) = F_n^{-1}(u)$, $F_n \to F$ on $\mathbb{R}^d$ and $F_n, n \geqslant 1$ is strictly increasing on $\mathbb{R}^d$
2. $\{X_n'(u)\}_{n\geqslant 1}$ be a sequence of random variables with $X_n'(u) = F_n^{'-1}(u)$, $F_n' \to F$ on $\mathbb{R}^d$ and $F_n, n \geqslant 1$ is strictly increasing on $\mathbb{R}^d$

Common random number simulation of X_n, X_n', means that for each iteration n, we simulate $U_n \sim Unif(0,1)$ and generate the random variables, $X_n(U_n), X_n'(U_n)$, from the same U_n.

Theorem 2.3. *Suppose that X_n, X_n' are defined as above and the limiting distribution F is a continuous, invertible mapping on $[0,1]$. If X_n, X_n' are simulated according to common random numbers, then $|X_n(U_n) - X_n'(U_n)| \to 0$ almost surely.*

Proof. First note that conditions 1 and 2 of Proposition 2.2 are satisfied by the definitions of X_n, X_n'.

Let $K_k = [\frac{1}{k}, 1 - \frac{1}{k}]$, $k > 2$ be a compact set. Then by definition of uniform convergence for any $\epsilon > 0$, there exists an $N_1, N_2 \in \mathbb{N}$ such that for any $n_1 \geqslant N_1$ and $n_2 \geqslant N_2$ and $u \in K_k$,

$$|F_{n_1}^{-1}(u) - F^{-1}(u)| \leqslant \epsilon/2$$
$$|F_{n_2}^{'-1}(u) - F^{-1}(u)| \leqslant \epsilon/2$$

So for $n \geqslant \max\{N_1, N_2\}$,

$$|F_n^{-1}(u) - F_n^{'-1}(u)| \leqslant |F_n^{-1}(u) - F^{-1}(u)| + |F_n^{'-1}(u) - F^{-1}(u)| \leqslant \epsilon/2 + \epsilon/2 = \epsilon$$

This means that

$$P(\lim_{n\to\infty} |F_n^{-1}(u) - F_n^{'-1}(u)| \leqslant \epsilon) \geqslant P(U \in K_k) = 1 - \frac{2}{k}$$

Since $k > 2$ is arbitrary, we set $k \to \infty$ and so,

$$P\left(\lim_{n\to\infty} |F_n^{-1}(u) - F_n^{'-1}(u)| \leqslant \epsilon\right) = P\left(\lim_{n\to\infty} |X_n(u) - X_n'(u)| \leqslant \epsilon\right) = 1$$

□

3 Necessary and sufficient conditions for CRN to generate the Wasserstein distance

Proposition 3.1 (Proposition 2.2 of [7] for $\theta \in \mathbb{R}$)**.** *Fix $x, y \in \mathbb{R}$ and let $X = f(\theta_U, x)$ and $Y = f(\theta_V, y)$ with measurable $f : \mathbb{R}^2 \to \mathbb{R}$ where $\theta_U = F_\theta^{-1}(U)$, $\theta_V = F_\theta^{-1}(V)$ and $U, V \sim Unif(0,1)$. Suppose for $z \in \{x, y\}$, $f(\cdot, z)$ is a non-decreasing continuous function with $E[X^2], E[Y^2] < \infty$. Then*

$$W(\mathcal{L}(X), \mathcal{L}(Y)) = \inf_{(X,Y)\sim J(X,Y)} E[(X-Y)^2] = E[(f(\theta_U, x) - f(\theta_U, y))^2] \tag{1}$$

or

$$\sup_{(U,V)\sim J(U,V)} E[f(\theta_U, x) f(\theta_V, y)] \tag{2}$$

is attained by setting $U = V$.

The fundamental problem with extending this to $\mathbb{R}^d, d > 1$ is that there is no practical ordering in $\mathbb{R}^d$. Both of the following claim that optimality cannot be extended to $\mathbb{R}^d$ in a practical way: Remark 2.30 and Chapter 2 of this

Theorem 3.2. *For fixed $x, y \in \mathcal{X}$, let $X = f(\theta_U, x)$ and $Y = f(\theta_V, y)$ be random functions with measurable $f : \mathbb{R}^2 \mapsto \mathbb{R}$ and $\theta_U = F_\theta^{-1}(U)$ and $\theta_V = F_\theta^{-1}(V)$ with $U, V \sim Unif(0,1)$. For $z \in \{x, y\}$, assume $\theta \mapsto f(\theta, z)$ is of bounded variation (see section 3.1 of [8] for definition) and has finite second moments.*

Let $I_z = \{\theta \mid f(\theta, z)$ is stochastically monotone$\}$ and $D_z = \{\theta \mid f(-\theta, z)$ is stochastically monotone$\}$ Define the set $A = (I_x \cap I_y) \cup (D_x \cap D_y)$.

- *If $P(A) = 1$,*

$$W_2(\mathcal{L}(X), \mathcal{L}(Y))^2 = E[(f(\theta_U, x) - f(\theta_U, y))^2] \tag{3}$$

  i.e. $(f(\theta_U, x) - f(\theta_U, y))^2$ is an unbiased estimator of $W_2(\mathcal{L}(X_1), \mathcal{L}(Y_1))^2$.

- *If $P(A) = 0$ (that is the function $f(\cdot, x)$ is increasing and $f(\cdot, y)$ is decreasing over the*

same intervals of positive measure or vice versa),

$$W_2(\mathcal{L}(X_1), \mathcal{L}(Y_1))^2 = E[(f(\theta_{1-U}, x) - f(\theta_U, y))^2] \tag{4}$$

i.e. $(f(\theta_{1-U}, x) - f(\theta_U, y))^2$ *is an unbiased estimator of* $W_2(\mathcal{L}(X_1), \mathcal{L}(Y_1))^2$.

- *If* $0 < P(A) < 1$ *(there are intervals of positive measure over which* θ *is increasing on* $f(\cdot, x)$ *and decreasing on* $f(\cdot, y)$ *or vice versa), then*

$$E[(f(\theta_U, x) - f(\theta_U, y))^2] - e \leqslant W_2(\mathcal{L}(X_1), \mathcal{L}(Y_1))^2 \leqslant E[(f(\theta_U, x) - f(\theta_U, y))^2] \tag{5}$$

where $e = 2E[(f(\theta_{1-U}, x) - f(\theta_U, x))f(\theta_U, y)1_{A^C}]$.

$$W_2(\mathcal{L}(X), \mathcal{L}(Y))^2 \coloneqq \inf_{(X,Y)\sim J(X,Y)} E[(X - Y)^2] \tag{6}$$

$$= E[X^2] + E[Y^2] - 2 \sup_{(X,Y)\sim J(X,Y)} E[XY] \tag{7}$$

$$= E[(F_X^{-1}(U) - F_Y^{-1}(U))^2] \tag{8}$$

Proof of Theorem 3.2. Fix x, y and denote $g(U) = f(F_\theta^{-1}(U), x)$ and $h(V) = f(F_\theta^{-1}(V), y)$. We write $J(U, V)$ to be the set of joint distributions such that $\mathcal{L}(U) = Unif(0, 1), \mathcal{L}(V) = Unif(0, 1)$. Finally, note that since $\theta = F^{-1}(U)$ where F is invertible, we interchangeably write the set A to signify A (defined on θ) and $F(A)$ (defined on U, V).

Also note that since $g(U), h(V)$ are assumed to be of bounded variation, the sets I_x, I_y, D_x, D_y can be written as the union of countable intervals by corollary 3.6 of [8].

First we show that

$$\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_A] = E[g(U)h(U)1_A] \tag{9}$$

Denote $I_x \cap I_y = \cup_{k\geqslant 1} J_k$ and $A\backslash(I_x \cap I_y) = \cup_{k\geqslant 1} L_k$ where J and L are intervals. We have

$$
\begin{aligned}
&\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_A] \\
&\leqslant \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{I_x\cap I_y}] + \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{A\backslash(I_x\cap I_y)}] \\
&= \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{I_x\cap I_y}] + \sup_{J(U,V)} E[(-g(U))(-h(V))1_{A\backslash(I_x\cap I_y)}] \\
&= \sup_{(U,V)\sim J(U,V)} E[\sum_{k\geqslant 1} g(U)h(V)1_{J_k}] + \sup_{(U,V)\sim J(U,V)} E[\sum_{k\geqslant 1} (-g(U))(-h(V))1_{L_k}] \\
&= \sup_{(U,V)\sim J(U,V)} \sum_{k\geqslant 1} E[g(U)h(V)1_{J_k}] + \sup_{(U,V)\sim J(U,V)} \sum_{k\geqslant 1} E[(-g(U))(-h(V))1_{L_k}] \\
&\leqslant \sum_{k\geqslant 1} \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{J_k}] + \sum_{k\geqslant 1} \sup_{(U,V)\sim J(U,V)} E[(-g(U))(-h(V))1_{L_k}] \\
&= \sum_{k\geqslant 1} E[g(U)h(U)1_{J_k}] + \sum_{k\geqslant 1} E[(-g(U))(-h(U))1_{L_k}] \\
&= E[g(U)h(U)1_{I_x\cap I_y}] + E[(-g(U))(-h(U))1_{A\backslash(I_x\cap I_y)}] \\
&= E[g(U)h(U)1_A]
\end{aligned}
$$

The third equality is by the Dominated Convergence Theorem, Theorem 1.5.8 of [5], and the second last equality is by Proposition 3.1.

Since $E[g(U)h(U)] \leqslant \sup_{J(U,V)} E[g(U)h(V)] \leqslant E[g(U)h(U)]$, equality follows. Note that Proposition 3.1 can still be applied even if $g(U)h(U)1_{J_k}$ and $g(U)h(U)1_{L_k}$ are no longer the product of right continuous functions, which is a result of the fact that 1_{J_k} may not necessarily be right continuous. This is because the theorem still applies by discussions in Section 4 of [3] that note that rather than requiring that $f(\theta, x)$ be right continuous it is simply necessary to assume that the discontinuous points are countable and have left and right limits. This is a necessary condition for bounded variation (assumed in the theorem). Since the function I_A has countably many discontinuities, Proposition 3.1 can be used.

Second we show that

$$\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{A^c}] = E[g(U)h(1-U)1_{A^c}] \tag{10}$$

Suppose A^c can also be written as countable intervals. The set A^c represents areas where either g is increasing and h is decreasing or vice versa. By similar reasoning to equation 9,

$$\begin{aligned}
\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{A^c}] &= -\inf_{(U,V)\sim J(U,V)} E[g(U)(-h(V))1_{A^c}] \\
&= -\inf_{(U,V)\sim J(U,V)} E[\sum_{I\in A^c} g(U)(-h(1-U))I] \\
&= -\inf_{(U,V)\sim J(U,V)} \sum_{I\in A^c} E[g(U)(-h(1-U))I] \\
&\leqslant -\sum_{I\in A^c} \inf_{(U,V)\sim J(U,V)} E[g(U)(-h(1-U))I] \\
&= -\sum_{I\in A^c} E[g(U)(-h(1-U))I] \\
&= -E[g(U)(-h(1-U))1_{A^c}] \\
&= E[g(U)h(1-U)1_{A^c}]
\end{aligned}$$

The third equality is by the Dominated Convergence Theorem, Theorem 1.5.8 of [5], and the third last equality is by Theorem 2 of [3].

Again, since $E[g(U)h(1-U)1_{A^c}] \leqslant \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{A^c}] \leqslant E[g(U)h(1-U)1_{A^c}]$, equality follows.

Case 1: Suppose $P(A) = 1$. We write $P(A^c) = P(U, V) \in F(A^c))$

$$\begin{aligned}
&\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)] \\
&= \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_A] + E[g(U)h(V)1_{A^c}] \\
&= \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_A] + E[g(U)h(V) \mid (U,V) \in F(A^c)]P(A^c) \\
&= \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_A] && \text{Since } P(A^c) = 0 \\
&= E[g(U)h(U)1_A] && \text{by equation 9} \\
&= E[g(U)h(U)]
\end{aligned}$$

By equation 6, equation 3 follows.

Case 2: Suppose $P(A) = 0$.

$$\begin{aligned}
&\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)] \\
&= E[g(U)h(1-U)1_{A^c}] && \text{by equation 10} \\
&= E[g(U)h(1-U)]
\end{aligned}$$

By equation 6, equation 4 follows.

Case 3: Suppose $0 < P(A) < 1$.

$$\begin{aligned}
&\sup_{(U,V)\sim J(U,V)} E[g(U)h(V)] \\
&\leqslant \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_A] + \sup_{(U,V)\sim J(U,V)} E[g(U)h(V)1_{A^c}] \\
&= E[g(U)h(U)1_A] + E[g(U)h(1-U)1_{A^c}] && \text{by equations 9 and 10}
\end{aligned}$$

By equation 6, equation 5 follows. □

Theorem 3.3. *Let $X_n, X'_n \in \mathbb{R}$ be two copies of a Markov chain that converge to a stationary distribution π such that $X_n = f(\theta_n, X_{n-1})$ where $\{\theta_n\}_{n\geqslant 1}$ are i.i.d. and the following conditions hold*

1. *$P(A) = 1$ in the notation of Theorem 3.2 are satisfied for all $x, x' \in \mathcal{X}$*
2. *the distance k is a quasi-antitone function (for example: $|x-y|^p, p \geqslant 1 f(x-y)$ where f is convexe) and is strictly convex*
3. *If $E[k(x,y)] \leqslant E[k(x',y')]$, then $E[k(f(\theta,x), f(\theta,y))] \leqslant E[k(f(\theta,x'), f(\theta,y'))]$*

Then the Wasserstein distance is attained when using common random numbers for $n \in \mathbb{N}$

$$W_k(\mathcal{L}(X_n), \mathcal{L}(X'_n)) = E[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x'))] \tag{11}$$

Proof. First note that since k is strictly convexe, by Theorem 2.9 of this,

$\inf_{\times_{i=1}^n \mathcal{J}(\theta_i,\theta'_i)} E[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}'_n, x'))]$ is attainable. That is, there exists $\times_{i=1}^n J(\theta_i, \theta'_i) \in \times_{i=1}^n \mathcal{J}(\theta_i, \theta'_i)$ such that

$$\inf_{\times_{i=1}^n \mathcal{J}(\theta_i,\theta'_i)} E[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}'_n, x'))] = E_{\times_{i=1}^n J(\theta_i,\theta'_i)}[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}'_n, x'))]$$

Next note that the infimum is bounded above by the expectation using CRN.

$$\inf_{\times_{i=1}^n \mathcal{J}(\theta_i,\theta'_i)} E[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}'_n, x'))] \leqslant E[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x'))]$$

We will prove (by contradiction) that equation 11 holds for $n = 2$.

Case1: Suppose that $\theta_1 \neq \theta_1'$ w.p.1.,

$$\begin{aligned}
&\inf_{\mathcal{J}(\theta_1,\theta_1')\times\mathcal{J}(\theta_2,\theta_2')} E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2', f(\theta_1', x)))\big] \\
&= \inf_{\mathcal{J}(\theta_2,\theta_2')} E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2', f(\theta_1', x)))\big] && \text{by assumption} \\
&= E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2, f(\theta_1', x)))\big] \\
&\geqslant E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2, f(\theta_1, x)))\big] && \text{by condition 2}
\end{aligned}$$

This is a contradiction, so $\theta_1 = \theta_1'$ w.p. 1.

Case 2: Suppose that $\theta_2 \neq \theta_2'$ w.p.1.,

$$\begin{aligned}
&\inf_{\mathcal{J}(\theta_1,\theta_1')\times\mathcal{J}(\theta_2,\theta_2')} E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2', f(\theta_1', x)))\big] \\
&= \inf_{\mathcal{J}(\theta_1,\theta_1')} E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2', f(\theta_1', x)))\big] && \text{by assumption} \\
&\geqslant \inf_{\mathcal{J}(\theta_1,\theta_1')} E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2, f(\theta_1', x)))\big] && \text{by condition 2} \\
&= E\big[k(f(\theta_2, f(\theta_1, x)), f(\theta_2, f(\theta_1, x)))\big]
\end{aligned}$$

This is a contradiction, so $\theta_2 = \theta_2'$ w.p. 1.

Must prove for $n > 2$ using induction. □

Lemma 3.4. *Let $X_n, X_n' \in \mathbb{R}$ be two copies of a Markov chain. If the conditions of Theorem 3.3 are satisfied and X_n is geometrically ergodic, then $X_n, X_n' \in \mathbb{R}$ generated using CRN will converge almost surely.*

$$P(\lim_{n\to\infty} k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x')) = 0) = 1 \qquad a.s.$$

Proof. If a Markov chain is geometrically ergodic, then there exists a π finite function $M(x)$ and a $\rho < 1$ such that for any $x \in \mathbb{R}$, $W_k(\mathcal{L}(X_n \mid X_0 = x), \pi) \leqslant M(x)\rho^n$. This means that for

starting values x, x',

$$W_k(\mathcal{L}(X_n \mid X_0 = x), \mathcal{L}(X_n \mid X_0 = x')) \leqslant 2\max\{M(x), M(x')\}\rho^n$$

Since the conditions of Theorem 3.3 are satisfied, $E\big[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x'))\big] = W_k(\mathcal{L}(X_n \mid X_0 = x), \mathcal{L}(X_n \mid X_0 = x'))$ and so,

$$E\big[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x'))\big] \leqslant 2\max\{M(x), M(x')\}\rho^n$$

Let $\epsilon > 0$

$$\begin{aligned}
\sum_{n=0}^{\infty} P(k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x')) > \epsilon) &\leqslant \sum_{n=0}^{\infty} \frac{E\big[k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x'))\big]}{\epsilon} \\
&\leqslant \sum_{n=0}^{\infty} \frac{2\max\{M(x), M(x')\}\rho^n}{\epsilon} \\
&= \frac{2\max\{M(x), M(x')\}}{\epsilon(1-\rho)} \\
&< \infty
\end{aligned}$$

By the Borel-Cantelli Lemma, $P(k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x')) > \epsilon \text{ i.o.}) = 0$, which is equivalent to saying that $P(\lim_{n\to\infty} k(f_n(\vec{\theta}_n, x), f_n(\vec{\theta}_n, x')) = 0) = 1$ and so $X_n, X_n' \in \mathbb{R}$ generated using CRN will converge almost surely. □

4 $W_{\|\cdot\|_1}(\mathcal{L}(X_n), \mathcal{L}(X)) \to 0 \iff X_n \xrightarrow{d} X$

Let $(\mathcal{X}, \mathcal{F}, \lambda)$ be a measure space with $\mathcal{F}$ as the Borel σ-algebra and equipped with a metric d so that $(\mathcal{X}, d)$ is a Polish metric space. We define two random variables X, Y on $(\mathcal{X}, \mathcal{F}, \lambda)$ where $\mathcal{L}(X)$ is the distribution of X. Denote $\mathcal{P}(\mathcal{X})$ as the set of probability measures on $(\mathcal{X}, \mathcal{F})$. For

$\mathcal{L}(X) = \pi, \mathcal{L}(Y) = \mu \in \mathcal{P}(\mathcal{X})$, define the set of couplings $J(X, Y)$ as

$$J(X,Y) = \{v \in \mathcal{P}(\mathcal{X} \times \mathcal{X}) : v(A_1 \times \mathcal{X}) = \pi(A_1), v(\mathcal{X} \times A_2) = \mu(A_2) \text{ for all } A_1, A_2 \in \mathcal{F}\}$$

The L^p-Wasserstein distance can then be defined as follows for $p \in [1, \infty)$

$$W_p(\pi, \mu) = \inf_{(X,Y)\sim J(X,Y)} E[d(X,Y)^p]^{1/p}$$

Total variation is defined as $||\pi - \nu||_{TV} = \sup_{A\in\mathcal{F}} |\pi(A) - \nu(A)|$.

Define a Markov chain $\{X_n\}_{n\geqslant 1}$ in $(\mathcal{X}, \mathcal{F})$ such that the density of X_n is denoted as f_n, and the distribution is denoted as ν_n. Suppose further that the Markov chain has a corresponding stationary distribution π.

The following theorem states that assuming dominated expectations, convergence in the Wasserstein distance is a necessary and sufficient condition for weak convergence.

Theorem 4.1. *Let $\{X_n\}_{n\geqslant 1}$ be a Markov chain such that there exists a random variable Y where $|X_n| \leqslant Y$ for all n and $E[Y] < \infty$. Then $X_n \Rightarrow X$ if and only if $W_1(\nu_n, \pi) \to 0$ where $\mathcal{L}(X_n) = \nu_n$ and $\mathcal{L}(X) = \pi$.*

Proof. $W_1(\nu_n, \pi) \to 0 \implies X_n \Rightarrow X$: If $W_1(\nu_n, \pi) \to 0$ then there exists random variables Z_n and Z where $Z_n \overset{d}{=} X_n$ and $Z \overset{d}{=} X$ such that $E[|Z_n - Z|] \to 0$. Since L^1 convergence implies weak convergence, $Z_n \Rightarrow Z$ and by equivalence $X_n \Rightarrow X$.

$X_n \Rightarrow X \implies W_1(\nu_n, \pi) \to 0$: If $X_n \Rightarrow X$ then there exists random variables Z_n such that $Z_n \overset{d}{=} X_n$ and $Z_n \to X$ almost surely (Theorem 3.2.2 of [5]). Since by assumption there exists a random variable Y where $|X_n| \leqslant Y$ for all n and $E[Y] < \infty$ then by equivalence $|Z_n| \leqslant Y$ for all n and so by the dominated convergence theorem (see Theorem 1.19 of [6]), $E[|Z_n - X|] \to 0$. And so, $W_1(\nu_n, \pi) \to 0$. □

This theorem is consistent with Theorem 6.9 of this, which is a more general statement.

The implications for common random number are the following:

Remark 2.19 (Translations). A nice feature of the Wasserstein distance over a Euclidean space $\mathcal{X} = \mathbb{R}^d$ for the ground cost $c(x,y) = \|x-y\|^2$ is that one can factor out translations; indeed, denoting $T_\tau : x \mapsto x - \tau$ the translation operator, one has

$$\mathcal{W}_2(T_{\tau\sharp}\alpha, T_{\tau'\sharp}\beta)^2 = \mathcal{W}_2(\alpha,\beta)^2 - 2\langle \tau-\tau', \mathbf{m}_\alpha - \mathbf{m}_\beta\rangle + \|\tau-\tau'\|^2,$$

where $\mathbf{m}_\alpha \overset{\text{def.}}{=} \int_{\mathcal{X}} x d\alpha(x) \in \mathbb{R}^d$ is the mean of α. In particular, this implies the nice decomposition of the distance as

$$\mathcal{W}_2(\alpha,\beta)^2 = \mathcal{W}_2(\tilde{\alpha},\tilde{\beta})^2 + \|\mathbf{m}_\alpha - \mathbf{m}_\beta\|^2,$$

where $(\tilde{\alpha},\tilde{\beta})$ are the "centered" zero mean measures $\tilde{\alpha} = T_{\mathbf{m}_\alpha\sharp}\alpha$.

Figure 2: https://arxiv.org/pdf/1803.00567. See Remark 2.31 for Wass distance between two Gaussians in R^d

- $E_{CRN}[|X_n - X'_n|] \to 0$ implies **show counter example** $W_1(\mathcal{L}(X_n), \mathcal{L}(X'_n)) \to 0$ implies $\mathcal{L}(X_n) \implies \mathcal{L}(X'_n)$
- $E[X_n] - E[X'_n] \nrightarrow 0$ implies $\mathcal{L}(X_n) \nRightarrow \mathcal{L}(X'_n)$ implies $W_1(\mathcal{L}(X_n), \mathcal{L}(X'_n)) \nrightarrow 0$

5 Summary

Theorem 5.1. *Let X_n be a sequence of random variables with distribution functions F_n. The following are equivalent for some random variable X with distribution function F*

1. $\mathcal{L}(X_n) \to \mathcal{L}(X)$ *ie. X_n converges in distribution to X_n.*
2. $W_d(\mathcal{L}(X_n), \mathcal{L}(X)) \to 0$
3. $E[d(F_n^{-1}(U), F^{-1}(U))] \to 0$ *if $d = f(x-y)$ and f is a convex function.*
4. $|F_n^{-1}(U) - F^{-1}(U)| \to 0$ *a.s.*

Can can we extend these theorems? We've got necessary and sufficient conditions under which $W_d(\mathcal{L}(X_n), \mathcal{L}(X)) = E[d(f_n(U), f(U))]$ when f_n, f are arbitrary and $U \in [0,1]$.

(3.1) *Definition.* Random variables $\boldsymbol{T} = (T_1, \cdots, T_n)$ are said to be associated if

$$\operatorname{Cov}\{f(\boldsymbol{T}), g(\boldsymbol{T})\} \geqq 0,$$

for all functions f and g which are non-decreasing in each argument and for which $E\{f(\boldsymbol{T})\}$, $E\{g(\boldsymbol{T})\}$, and $E\{f(\boldsymbol{T})g(\boldsymbol{T})\}$ exist.

(3.2) *Property.* Any subset of associated random variables is associated.

(3.3) *Property.* If two sets of associated random variables are independent of one another, then their union is a set of associated random variables.

(3.4) *Property.* The set consisting of a single random variable is associated.

(3.5) *Property.* Non-decreasing functions of associated random variables are associated.

A class of processes for which non-negative correlation can be guaranteed is stochastically monotone Markov chains (SMMC). This class was introduced by

Figure 3: This is perhaps my only hope of extending 3.2 to $\mathbb{R}^d$. This paper says that when using CRN, variance is reduced. Could one somehow define functions to show that CRN actually generates the optimal reduction in covariance? Also, what does it mean for f to be non-decreasing in each argument? (Comparing Stochastic Systems Using Regenerative Simulation with Common Random Numbers Philip Heidelberger, Donald L. Iglehart https://www.jstor.org/stable/1426860?seq=7)

Theorem 2. *Let X and Y be random variables with distribution functions F and G respectively and joint distribution function H, and let $k(x, y)$ be quasi-monotone and right continuous. If the expectations $\mathscr{E}_{H_-} k(X, Y)$ and $\mathscr{E}_{H_+} k(X, Y)$ exist (even if infinite valued), then the set of values of $\mathscr{E}_H k(X, Y)$ when H ranges over $\mathscr{H}(F, G)$ is the (possibly unbounded) closed interval $[\mathscr{E}_{H_-} k(X, Y), \mathscr{E}_{H_+} k(X, Y)]$ when either of the following is satisfied:*

(a) *$k(x, y)$ is symmetric and* (i) *holds (in this case $-\infty \leqq \mathscr{E}_{H_-} k(X, Y) \leqq \mathscr{E}_{H_+} k(X, Y) < +\infty$),*

(b) *for some x_0 and y_0,* (ii) *holds and at least one of $\mathscr{E}_{H_+} k(X, Y)$ and $\mathscr{E}_{H_-} k(X, Y)$ is finite.*

Thus under the assumptions of Theorem 2 the infimum and the supremum of $\mathscr{E}_H k(X, Y)$ for $H \in \mathscr{H}(F, G)$ are achieved by H_- and H_+ respectively and they are given by

$$\mathscr{E}_{H_+} k(X, Y) = \int_0^1 k(F^{-1}(u), G^{-1}(u))\, du,$$

$$\mathscr{E}_{H_-} k(X, Y) = \int_0^1 k(F^{-1}(u), G^{-1}(1-u))\, du.$$

Of course if k is quasi-antitone the infimum is achieved by H_+ and the supremum by H_-.

Figure 4: Something powerful is noted here: for any $H \in \mathcal{H}(\mathcal{L}(X), \mathcal{L}(Y))$, $E_H[k(X,Y)] \in [E_{H_+[X,Y]}, E_{H_-[X,Y]}]$. This set is convex and compact, but again only applies to $\mathbb{R}$

References

[1] Erich Barvínek, Ivan Daler, and Jan Franců. "Convergence of sequence of inverse functions". In: *Arch. Math.* 27 (Jan. 1991).

[2] Patrick Billingsley. *Probability and Measure: Anniversary Edition*. New York: Wiley Series in Probability and Statistics, 2012.

[3] Stamatis Cambanis, Gordon Simons, and William Stout. "Inequalities for Ek(X, Y) when the marginals are fixed". In: *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* 36.4 (1976), pp. 285–294. DOI: 10.1007/BF00532695.

[4] Persi Diaconis and David Freedman. "Iterated random functions". In: *SIAM Review* 41.1 (1999), pp. 45–76. DOI: 10.1137/S0036144598338446.

[5] Rick Durrett. *Probability: Theory and Examples*. 4th ed. New York: Cambridge Series in Statistical and Probabilistic Mathematics, 2010. DOI: 10.5555/1869916.

[6] Lawrence C. Evans and Ronald F. Gariepy. *Measure Theory and Fine Properties of Functions*. 1st ed. CRC Press, 2015. DOI: 10.1201/b18333.

[7] Paul Glasserman and David D. Yao. "Some guidelines and guarantees for common random numbers". In: *Management Science* 38 (1992), pp. 884–908. DOI: 10.5555/2909967.2909978.

[8] Elias M Stein and Rami Shakarchi. *Real analysis: measure theory, integration, and Hilbert spaces*. Princeton lectures in analysis. Princeton, NJ: Princeton Univ. Press, 2005. DOI: 10.2307/j.ctvd58v18.

[9] David Steinsaltz. "Locally Contractive Iterated Function Systems". In: *Annals of Probability* 27.4 (1999), pp. 1952–1979. DOI: 10.1214/aop/1022874823.

[10] Örjan Stenflo. "A Survey of Average Contractive Iterated Function Systems". In: *Journal of Difference Equations and Applications* 18.8 (2012), pp. 1355–1380. DOI: 10.1080/10236198.2011.610793.